

Week 7 - Dry and Moist Air: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: $R_v = 461.5 \text{ J kg}^{-1} \text{ K}^{-1}$, $\epsilon = R_d/R_v \approx 0.622$, $L_v \approx 2.5 \times 10^6 \text{ J kg}^{-1}$. A useful estimate for lifting-condensation-level height is $z_{LCL} \approx 125(T - T_d)$ metres when T and T_d are in $^{\circ}\text{C}$. Approximate saturation vapour pressure over liquid water:

T ($^{\circ}\text{C}$)	-10	0	5	10	15	20	25
e_s (hPa)	2.87	6.11	8.72	12.27	17.04	23.37	31.67

YOU SHOULD TRY Question 1: Phase changes and atmospheric energy

For evaporation, condensation, melting, freezing, sublimation and deposition:

1. state the direction of the phase change;
2. state whether latent heat is absorbed from or released to the surroundings;
3. identify which two of these processes are especially important when cloud water or ice forms from water vapour.

IN CLASS Question 2: Relative humidity and dew point

At 25°C , an air parcel has water-vapour partial pressure $e = 17.0 \text{ hPa}$. Relative humidity is defined by

$$RH = 100 \frac{e}{e_s(T)},$$

where $e_s(T)$ is the saturation vapour pressure at the air temperature.

1. Calculate the relative humidity.
2. Use the table to estimate the dew-point temperature.
3. Explain physically what would happen if the parcel were cooled below its dew point while its water-vapour content initially remained unchanged.

EXTRA CHALLENGE Question 3: Water-vapour density

For the air in Question 2, calculate the density of water vapour using

$$e = \rho_v R_v T.$$

IN CLASS Question 4: Mixing ratio and specific humidity

The **water-vapour mixing ratio** is the mass of water vapour per unit mass of dry air,

$$r_v = \frac{m_w}{m_d},$$

whereas the **specific humidity** is the mass of water vapour per unit mass of moist air,

$$q = \frac{m_w}{m_w + m_d}.$$

At total pressure $p = 1000$ hPa and vapour pressure $e = 17.0$ hPa:

1. Calculate the mixing ratio using

$$r_v = \epsilon \frac{e}{p - e}.$$

2. Starting from the definitions above, show that

$$q = \frac{r_v}{1 + r_v},$$

and calculate q .

3. Give both r_v and q in g kg^{-1} and explain why they are numerically similar for ordinary atmospheric air.

YOU SHOULD TRY Question 5: Why moist air is less dense than dry air

A mixture contains 98% dry air and 2% water vapour by mole fraction. Take molar masses $M_d = 28.97 \text{ g mol}^{-1}$ and $M_v = 18.02 \text{ g mol}^{-1}$.

1. Calculate the mean molar mass of the mixture.
2. At the same pressure and temperature, would this mixture be denser or less dense than dry air? Explain.

IN CLASS Question 6: Estimating cloud-base height

For an unsaturated parcel lifted approximately adiabatically, take the parcel temperature to decrease at the dry-adiabatic lapse rate,

$$T(z) = T_0 - \Gamma_d z, \quad \Gamma_d \approx 9.8 \text{ K km}^{-1}.$$

The parcel dew-point temperature also decreases as the parcel expands and its vapour pressure falls; a useful approximation is

$$T_d(z) = T_{d0} - \Gamma_{dew} z, \quad \Gamma_{dew} \approx 1.8 \text{ K km}^{-1}.$$

The lifting condensation level (LCL) is reached when $T = T_d$.

For surface temperature $T_0 = 22^\circ\text{C}$ and dew point $T_{d0} = 14^\circ\text{C}$:

1. Show that

$$z_{LCL} \approx \frac{T_0 - T_{d0}}{\Gamma_d - \Gamma_{dew}},$$

and hence recover the familiar approximation $z_{LCL} \approx 125(T_0 - T_{d0})$ m when temperatures are in $^\circ\text{C}$.

2. Estimate the LCL height.
3. Explain physically why a larger surface $T_0 - T_{d0}$ difference usually implies a higher cloud base for a lifted parcel.

EXTRA CHALLENGE Question 7: A psychrometer

A ventilated psychrometer has a dry-bulb thermometer and a wet-bulb thermometer.

1. Why is the wet bulb usually colder than the dry bulb when the air is unsaturated?
2. What happens to the wet-bulb depression as relative humidity approaches 100%?
3. Why is ventilation important?

YOU SHOULD TRY Question 8: Latent heating by condensation

A rising cloud parcel condenses 2.0 kg of water. Estimate the latent heat released. If that energy were used only to warm 1000 kg of air with $c_p = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$, what temperature increase would it correspond to?